KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2024
Department of Computer Science and Engineering
CSE 4111
Machine Learning

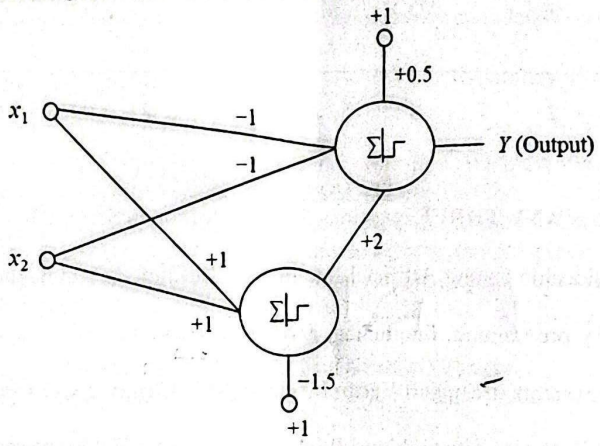
TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) "In Machine Learning (ML), explicit programming is not performed to solve a particular problem" – justify this statement. (08) [CO1]
   b) Briefly describe the basic learning types. Arrange the learning types in order of the increasing feedback from the environment; provide justification of the ordering. (09) [CO2]
   c) Differentiate learning and generalization in the context of ML. Which one mostly indicates the performance of any ML model and what techniques are employed to achieve the indicative? (09) [CO3]
   d) Does the following neural architecture solve 2 input non-linear problem? Justify your answer considering the inputs (0,0), (0,1), (1,0), and (1,1) at x_1 and x_2. (09) [CO3]



2. a) Describe importance of bias weight value in artificial neuron construction. "Changing only the bias weight value can change the operation of a neuron" – justify the statement for appropriate logic gates. (08) [CO1]
   b) What is the condition to use a function as activation function in artificial neural network? Graphically represents two different forms sigmoidal functions and their derivatives. Explain similarity and dissimilarity between the functions. (09) [CO2]
   c) What is the local gradient in Back Propagation (BP) learning algorithm? "Local gradient may be zero even error at neural network remain" – justify the statement with an appropriate activation function. (09) [CO2]
   d) Draw diagram of RBF networks with approximate parameter values to solve AND, OR, and XOR logic operations for two inputs. (09) [CO4]

3. a) What are the various ways to produce diverse networks for ensemble construction? Which one is more effective? Why? (07) [CO1]
   b) Briefly describe independent, sequential, and simultaneous training of component networks in ensemble construction. Mention prominent ensemble methods for different ensemble training methods. (10) [CO2]
   c) Consider two computation systems: (i) multiple processing units and (ii) single high power unit. What are the preferable ensemble methods for the systems individually and why? (10) [CO4]
   d) Identify the learnable parameters for the below Convolutional Neural Network (CNN) to classify numeral images having size 28×28 pixels. Also, count the learnable parameters. (08) [CO3]

$$I_{28\times28} \rightarrow \{6K1_{5\times5}C1_{24\times24} - S_{2\times2}6S1_{12\times12}\} \rightarrow \{6K2_{5\times5}C2_{8\times8} - S_{2\times2}6S2_{4\times4}\} \rightarrow W_{o96\times10} \rightarrow O_{10}$$

where symbols hold usual meanings.

CamScanner

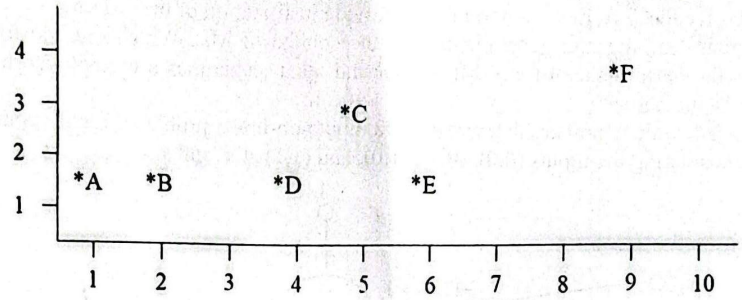
4. a) Discuss advantages and disadvantages of Decision Trees relative to Neural Networks for classification on the basis of important issues (e.g., data type, compatibilities, sample complexities, model interpretabilities). (10) [C

b) Consider the following table where attribute "Play Tennis" represents class label. Construct a Decision Tree using the table measuring information gain and then generate rules from the Decision Tree. (10) [CO3]

Sl. No.	Outlook	Temp	Humidities	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Rain	Mild	High	Weak	Yes
4	Rain	Cold	Normal	Weak	Yes
5	Rain	Cold	Normal	Strong	No
6	Sunny	Mild	Normal	Strong	Yes

c) Briefly discuss strength and weakness of K-means clustering. (07) [CO2]

d) Draw a dendrogram considering six data points shown below; no need exact distance calculation. (08) [CO3]



SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Explain the relationship among Artificial Intelligence, Machine Learning, and Deep Learning. (06) [CO1]

b) What is meant by pre-training, fine-tuning, and transfer learning? Give example of each case. (09) [CO1]

c) What are support vectors in Support Vector Machine (SVM)? How does Linear SVM work? (12) [CO2]

d) What is curse of dimensionality problem? Why is it an issue in ML? Briefly explain a dimensionality reduction technique. (08) [CO2]

6. a) Differentiate between heuristic and meta-heuristic with examples. (06) [CO1]

b) "Meta-heuristics are designed to solve a problem more quickly and efficiently than classic methods" – justify it. (10) [CO2]

c) Explain exploration and exploitation with suitable example. (10) [CO1]

d) "Crossover and mutation are not always beneficial" – is it true? If so, justify with a numeric example. (09) [CO2]

7. a) Write simple Genetic Algorithm (GA). Why is choosing proper fitness function important in GA? Explain with example. (10) [CO2]

b) Suppose, you want to minimize $f(x) = x^2$ where $-5 \leq x \leq 5$ using GA. Assume the required initial values, and consider MSB as sign bit and rest of the bits are magnitude during encoding. Now, perform calculation for one generation by hand. (12) [CO4]

c) Write an algorithm for Enhanced Edge Recombination Operator with proper example. (13) [CO3]

8. a) What is Swarm? Write properties of a Swarm Intelligence (SI) system. (05) [CO1]

b) Explain Particle Swarm Optimization (PSO) technique. What are the drawbacks of this technique? (12) [CO2]

c) Explain Ant Colony Optimization (ACO) to solve Traveling Salesman Problem (TSP). (10) [CO4]

d) Which bees perform exploration and exploitation in Artificial Bee Colony (ABC) optimization? What is significance of parameter "limit" in ABC optimization? (08) [CO2]

CamScanner

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2023
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the immediate right column of the questions indicate full marks.
iii) The rightmost column indicates course outcomes.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) Define machine learning and discuss its different types. (06) [CO1]
   b) Derive the Lagrangian dual objective function and formulate the equations that uniquely characterize the solution to both the primal and dual problems for the Support Vector Machine (SVM). (14) [CO3]
   c) Explain how Directed Acyclic Graph (DAG) SVM addresses the drawback of the 1-vs-1 SVM approach, where the number of SVMs increases super-linearly with number of classes, in data classification. (08) [CO2]
   d) Discuss the relationship between Gini Impurity and Information Gain. (07) [CO2]

2. a) How does a separating hyperplane that perfectly classifies all of the training observations lead to sensitivity to individual observations? Explain in detail. (10) [CO2]
   b) Can SVM be over trained? What can be the effect of over training? How can over-fitting be detected for a SVM model? (07) [CO2]
   c) Consider the following two hyperplanes: (08) [CO3]

$$5 + 2x_1 + 5x_2 \quad \text{(for classifier 1)}$$
$$5 + 20x_1 + 50x_2 \quad \text{(for classifier 2)}$$

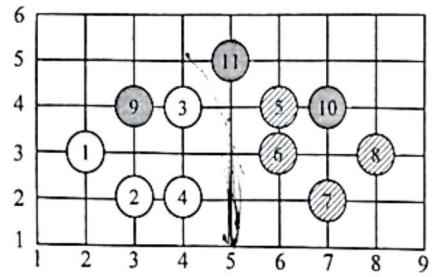
Compute the distance error function and identify which classifier built with these hyperplanes will have better performance.
   d) 'The maximal margin hyperplane for SVM depends directly on only a small subset of the observation' – justify the statement. (10) [CO1]

3. a) Explain Occam's razor principle. In the context of decision tree learning, when we prioritize simplicity over complexity, do you believe Occam's razor adequately justifies the algorithm's inclination toward simpler hypotheses? (10) [CO1]
   b) Discuss the impact of outliers on the performance of a maximum margin classifier. How can the classifier be sensitive to outliers and what techniques can be employed to address this issue? (12) [CO2]
   c) Consider the following table where attribute 'Status' represents class label. Find the root attribute of the decision tree using Gain Ratio as the attribute selection measure. (13) [CO3]

Department	Age	Status	Salary
Sales	31–35	Senior	Medium
Sales	26–30	Junior	Low
Marketing	36–50	Senior	High
Secretary	26–30	Senior	Medium
Secretary	36–50	Senior	High
Sales	36–50	Junior	High
Marketing	31–35	Senior	Medium
Marketing	26–30	Junior	Low
Secretary	31–35	Junior	Medium
Secretary	26–30	Junior	Low

4. a) Discuss the empty space phenomenon and demonstrate the working principle of kernel Principle Component Analysis (PCA) in dimensionality reduction for linearly inseparable input data. Under what conditions does kernel PCA function similarly to standard PCA? (11) [CO1]
   b) What is inductive bias in decision tree learning and which type of inductive bias is used in BFS-ID3 algorithm? Illustrate how BFS-ID3 algorithm exhibits the bias 'shorter trees are preferred over longer trees' with detailed explanation. (12) [CO2]

c) Consider the following training data: (12) [CO3]



Let $y_A = -1, y_B = +1$ be the class indicators for both classes where

$A = \{(1), (2), (3), (4)\}$ and $B = \{(5), (6), (7), (8)\}$

i) Just using the above standing plot, specify which of the points should be identified as support vectors.

ii) Draw the maximum margin line which separates the classes (you don't have to do any computations here). Write down the normalized normal vector $w \in \mathbb{R}^2$ of the separating line and the offset parameter $b \in \mathbb{R}$.

iii) Consider the decision rule: $H(x) =< w, x > +b$. Now determine the class for the points (9), (10) and (11).

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Suppose, we want to fit a line $y = mx + b$ through the points $(1, 2), (2, 3), (3, 4), (4, 5)$ and $(5, 6)$. Assume, initial values of m and b are 0.1 and 0.2 respectively. Calculate the initial error using mean squared error loss function and then apply a gradient descent step. Use learning rate, $\alpha = 0.01$ and calculate the error of fit again after the gradient descent step. (12) [CO3]

b) Demonstrate automatic differentiation (AD) using a suitable example. How does AD differ from numerical differentiation? (07) [CO1]

c) Explain why Convolutional Neural Network is preferred over Multi-Layer Perceptron for image related tasks. (04) [CO2]

d) Explain how artificial neural networks can be used to solve following problems. (12) [CO4]

i) Duplicate/ near duplicate image identification.

ii) Anomaly detection.

iii) Dimensionality reduction.

6. a) Briefly explain the idea of transfer learning. (05) [CO1]

b) What is exploration and exploitation in swarm intelligence algorithms? Explain how these features are provided in following algorithms. (16) [CO1]

i) Genetic algorithm (GA).

ii) Particle swarm optimization (PSO).

iii) Ant colony optimization (ACO).

iv) Artificial bee colony (ABC).

c) What are the disadvantages of Roulette Wheel selection? (03) [CO3]

d) Create a child using Cycle crossover and another child using Enhanced Edge Recombination for the following parents. (11) [CO3]

Parent 1	1	2	3	4	5	6

Parent 2	5	6	1	3	4	2

7. a) "PSO cannot be used for discrete optimization problem" – justify this statement. (05) [CO1]

b) Perform one iteration of PSO to minimize the function $f(x, y) = (x - 2)^2 + (y - 3)^2$ in the range $-10 \leq x, y \leq 10$. Parameters and initial values are given below where the symbols have their usual meanings. (10) [CO3]

Parameters:

$N = 2, \omega = 0.5,$
$C_1 = 1.5, C_2 = 1.5$
$r_1 = 0.3, r_2 = 0.7$

Initial values:

Particle 1: Position $(x_1, y_1) = (5, 5)$,
Velocity $(v_{1x}, v_{1y}) = (0.1, 0.1)$.
Particle 2: Position $(x_2, y_2) = (3, 1)$,
Velocity $(v_{2x}, v_{2y}) = (0.2, 0.2)$.

c) Write down the equations of transition probability and pheromone update of ACO. Also, describe the effects of key parameters. (10) [CO1]

d) Explain the employee bee phase, on looker bee phase and scout bee phase in ABC optimization algorithm. (10) [CO1]

CamScanner

8. a) Explain how clustering can be used for image color quantization. (04) [CO3]
   b) Explain how to calculate and interpret Silhouette score. (06) [CO2]
   c) What is the problem of random centroid initialization in case of k-means clustering? Explain how k-means++ or GA can be used to solve this problem (explain both approaches). (11) [CO2]
   d) Using agglomerative hierarchical clustering create a dendrogram for the data shown in following table. (14) [CO3]

Student_ID	Marks
1	15
2	12
3	33
4	25
5	40

CamScanner

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2021
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What do you mean by "Machine" and "Learning"? What are their individual features? In what prospects both the terms combine in Machine Learning (ML) subject? (07)
   b) "Training appropriate neural network architecture with appropriate training set is necessary to develop an appropriate model for a given task" – Justify the statement for a logic gate function case. (08)
   c) Draw the minimal neural architectures for the following three input (A, B and C) logic gate operations. (12)
      i) $NN1 = (AB)'C$ ii) $NN2 = ABC'$
      iii) $NN3 = (A + B)'C$ iv) $NN4 = (A + B)C'$
   d) Differentiate learning and generalization in the context of machine learning. Which one is more desirable and why? What are the measuring techniques of those? (08)

2. a) Why is a single neuron unable to solve XOR problem? Solve XOR problem with Radial Basis Function (RBF) network considering cluster centers (0, 1) and (1, 0) with appropriate radius of each center. (08)
   b) Write down the roles of activation function in artificial neural network? Compare two different forms of sigmoidal functions through graphical representations of Input vs Output and derivatives. (08)
   c) What is the naming significance of "Back Propagation (BP)" algorithm? What is the effect of learning rate in BP? How momentum parameter speedup training in BP? (07)
   d) What is the goal of neural network ensemble construction? Explain an ensemble method for enough data and an ensemble method for limited data. (12)

3. a) Why non-deterministic operation is preferable over deterministic operation in population-based optimization algorithms? How non-deterministic feature assures in Genetic Algorithm (GA), Particle Swarm Optimization (PSO) and Ant Colony Optimization (ACO)? (12)
   b) Explain velocity and position updates of particles in PSO mentioning corresponding equations in continuous function space. How the terms may transfer to solve Traveling Salesman Problem (TSP)? (09)
   c) Why both exploration and exploitation operations are necessary in an algorithm for better performance? (06)
   d) How individual parameters of PSO velocity update equation influence exploration and exploitation? (08)

4. a) What do you understand by the search terms Global Search, Local Search and Blind Search? Compare GA with a prominent Swarm Intelligence (SI) method on the basis of the search methods. (13)
   b) Explain the working principle of GA with a flowchart. Briefly explain importance of different operators and parameters in GA. (08)
   c) Suppose two TSP solutions are 1 – 2 – 4 – 3 – 6 – 5 and 1 – 5 – 3 – 4 – 2 – 6 where numeric values are city indexes. Generate an offspring from the solutions using Enhanced Edge Recombination algorithm. (08)
   d) Explain the fundamental differences between ACO and PSO to establish interaction among individuals. (06)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is 'K' in the K Nearest Neighbor (KNN) algorithm and why is the odd value of 'K' preferred over even values in the KNN algorithm? (05)

   b) We can easily find K Nearest Neighbors using Euclidean distance and then classify an instance. Then, why do we need K Dimensional tree (K-D Tree) for KNN? (05)

   c) Suppose, a satellite image can be segmented using four categories: water body, grass land, bare land and semi grassy area. Choose a suitable algorithm to segment the image and explain the process. (05)

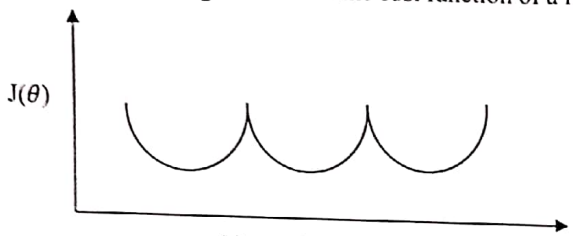
   d) Consider the following cost function and hypothesis for linear regression. Derive the normal equation, $\theta = (X^T X)^{-1} X^T Y$ to get the values of vector θ. (15)

$$j(\theta) = \frac{1}{2m}\sum_{i=1}^{m}\left(h_\theta\left(x^{(i)}\right) - y^{(i)}\right)^2$$

$$h_\theta\left(x^{(i)}\right) = \theta_0 + \theta_1 x_1^{(i)} + \cdots + \theta_n x_n^{(i)}$$

   How do you calculate θ when $X^T X$ is non-invertible?

   e) Consider the following outcome of the cost function of a linear regression model. (05)



   Is there any issue with the outcome? Explain the issue and solution (if any).

6. a) The cost function of logistic regression is given below: (15)

$$J(\theta) = -\frac{1}{m}\sum_{i=1}^{m}\left[y^{(i)}\log\left(h_\theta\left(x^{(i)}\right)\right) + \left(1 - y^{(i)}\right)\log\left(1 - h_\theta\left(x^{(i)}\right)\right)\right]$$

   Prove that, $\frac{\partial}{\partial\theta_j}J(\theta) = \frac{1}{m}\sum_{i=1}^{m}\left[h_\theta\left(x^{(i)}\right) - y^{(i)}\right]x_j^{(i)}$.

   b) Is there any similarity between a logistic regression model and a single neuron of a neural network (with sigmoid activation function)? Explain (if any). (07)

   c) What do you mean by overfitting and underfitting? Write down a way to minimize both the issues. (07)

   d) Suppose, one of your friends is planning to build a Machine Learning (ML) model to predict the diseases of many patients from various symptoms. He wants to sell the product to several hospitals. He has only 400 instances of data from 300 different patients. The dataset has both categorical and continuous features. There are only 5 diseases for prediction. Which ML model would you suggest to your friend and why? (06)

7. a) Consider the following dataset where attribute 'Salary' represents class label. (14)

Department	Age	Status	Salary
Sales	25..30	Junior	Low
Sales	31..35	Senior	Medium
Marketing	31..35	Senior	High
Admin	25..30	Junior	Medium
Admin	31..35	Senior	High
Admin	36..40	Senior	High
Marketing	36..40	Senior	High
Sales	25..30	Junior	Low
Marketing	36..40	Senior	Medium
Marketing	25..30	Junior	Low

   Find the root attribute of the Decision Tree (created from the dataset) using Gain Ratio as the attribute selection measure.

b) Consider the following problem. (14)

Minimize $f(x, y) = (x - 1)^2 + (y - 3)^2$

Subject to $x + y \leq 2, y \geq x$

Use the following formula for the minimization problem and find the minimum values for x and y.

Lagrangian, $L = f(x) + \sum_{i=1}^{p} \lambda_i \cdot h_i(x)$.

KKT constraints are as follows: $\frac{\partial L}{\partial x_i} = 0, i = 1,2, \cdots\cdots, d.$

$\lambda_i \geq 0, i = 1,2, \cdots\cdots, p.$

$h_i(x) \leq 0, i = 1,2, \cdots\cdots, p.$

$\lambda_i \cdot h_i(x) = 0, i = 1,2, \cdots\cdots, p.$

c) The learning task in Support Vector Machine (SVM) can be formulated as the following constrained optimization problem. (07)

Minimize $\mu(W, b)$

Subject to $y_i(W \cdot x_i + b) \geq 1, i = 1,2,3, \cdots\cdots, n.$

Write down the Lagrangian and KKT constraints for this problem.

8. a) Consider the case of a binary classification with a training data of 6 tuples as shown in the following table. (12)

X_1	X_2	Y	λ_i
0.38	0.48	+	65.52
0.59	0.71	−	65.52
0.82	0.31	−	65.52
0.75	0.90	−	0
0.19	0.59	+	0
0.41	0.35	+	0

The lagrange multipliers λ_i for each training tuple are solved using quadratic programming and the first three tuples are support vectors. Determine W and b to find the Maximum Margin Hyperplane of linear SVM.

b) Briefly explain the curse of dimensionality problem. (06)

c) Consider the following dataset with 2 dimensions. (17)

Dimension 1	Dimension 2
2.5	2.4
2.2	2.9
1.9	3.0
2.7	2.5
2.0	1.6
2.1	1.1
1.0	1.8
1.9	2.7

Using Principal Component Analysis (PCA), reduce the dataset to one dimension. Explain each step with necessary calculations.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2020
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 1.5 hours

FULL MARKS: 120

N.B. i) Answer **ANY TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY TWO** questions from this section in Script A)

1. a) Differentiate learning and generalization in the contexts of machine learning. Which one is more desirable and why? (08)

   b) Define classification, regression and logistic regression in short. Differentiate between the concepts: classification and regression; and regression and logistic regression. (10)

   c) Using K-Means clustering, cluster the following objects for $K = 2$.

Object	Attribute 1	Attribute 2
A	10	150
B	150	10
C	X	Y
D	150	150

   The attribute values of object C are $X =$ $Y =$ (Last three digits of your Roll Number). (12)

2. a) Suppose bias weight of an artificial neuron is $-X$, where X is the last three digits of your Roll Number. Sketch two input AND logic and OR logic by setting appropriate values of other weights of the neurons. (10)

   b) What are the significant properties of XOR in the context of machine learning? Is it possible to construct XOR logic with a neuron taking input from developed AND logic and OR logic? Do it if possible; otherwise explain the reasons for inability. (10)

   c) What is the local gradient in Back Propagation (BP) algorithm? "Local gradient may be zero even error remains" – Justify the statement with an appropriate activation function. (10)

3. a) Why dimensionality reduction is important in machine learning? How does principal component analysis reduce dimensionality? (10)

   b) What is Bayes theorem? Shortly describe Naive Bayes Classifier. (10)

   c) Consider the label data points in the following table. (10)

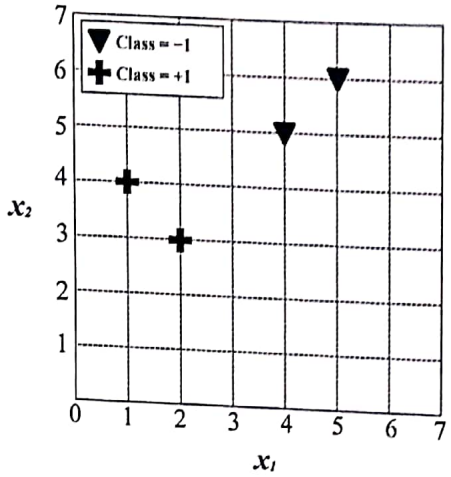
X_1	X_2	Label
05	15	+
10	30	−
30	40	+
40	100	−
80	130	+
120	250	−
Y_1	Y_2	?

   Identify the label of data point (Y_1, Y_2) using KNN for $K = 3$, where Y_1 = Last three digits of your Roll Number and $Y_2 = 2Y_1$.

SECTION B

(Answer **ANY TWO** questions from this section in Script B)

4. a) Suppose you are training on a tiny dataset with 4 points shown in the following figure. This dataset consists of two examples with class label +1 (denoted with plus), and two examples with class label −1 (denoted with triangle). (10)



   i. Find the weight vector w and bias b. What is the equation corresponding to the decision boundary?
   ii. Circle the support vectors and draw the decision boundary.

   b) Consider the following training dataset, where A, B, and C are three binary attributes, and D is a binary class label. (20)

Instance	A	B	C	D
1	0	0	0	0
2	0	0	1	0
3	0	1	0	1
4	0	1	1	1
5	1	0	1	1
6	1	0	1	1
7	1	1	0	0
8	1	1	0	0

   i. What is the expected information (entropy) required to classify the training dataset?
   ii. Using ID3 algorithm, build a decision tree for this classification problem.
   iii. Use the built decision tree to classify two samples below.

Instance	A	B	C	D
9	1	1	1	?
10	1	0	0	?

5. a) "In PSO, each and every particle changes position in each iteration even the particle is the global best (G)" – do you agree with the statement? Justify your answer with appropriate assumption in velocity update equation. (09)

   b) Is it possible to find out the population size from the Roulette wheel given below? Explain. (07)



   c) Apply the Enhanced Edge Recommendation algorithm on the following two chromosomes to produce two children. Show all the steps. (14)

Parent 1	4	2	1	3	5	6
Parent 2	4	3	1	2	6	5

6. a) "Pheromone evaporation pushes ACO convergence" – Is it true? Justify your opinion. (10)

b) What two requirements should a problem satisfy in order to be suitable for solving it by GA? (05)

c) Assume 4 cities {A, B, C, D}, which are represented by a fully connected graph. The following tables represent the pheromone levels on each edge of the graph and the distances between each city (assume the pheromone levels and distances are symmetric). (15)

Table-1

Pheromone Levels				
	A	B	C	D
A				
B	0.25			
C	0.11	0.98		
D	0.34	0.54	0.67	

Table-2

Distance				
	A	B	C	D
A				
B	12			
C	10	6		
D	8	15	3	

Assume an ant started its journey at city A and has travelled to city C.

i. What are the probabilities that the ant will travel to cities A, B and D? [Let $\alpha = 1, \beta = 1$]

ii. Assume the ant completes its tour using the route ABCD. What will be the pheromone levels on each edge once they have been updated? [Let Q = 100 and evaporation parameter = 0.5]

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2019
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours

FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What do you mean by learning? In biological system, what are the elements and how they response for learning? (08)
   b) Draw computational model of a single neuron. Solve different binary logics with the model. (09)
   c) What are the different types of memory? What are their significances and uses? (09)
   d) Calculate the weight set (W) of a Hetero-Associative memory for the following samples: (09)

Pattern, p	Source s(p)	Target, t(p)
1	1 0 0 0	1, 0
2	1 1 0 0	1, 0
3	0 0 1 1	0, 1
4	0 0 0 1	0, 1

   Finally verify the system with appropriate samples.

2. a) What do you understand by linear problem and non-linear problem in the context of machine learning? Why solving non-linear problem is challenging? Briefly explain with an appropriate example. (07)
   b) "An additional dimension (input) may transform a non-linear problem to linearly separable"- justify the statement with proper example. (08)
   c) Why training may not perform after certain iteration even though error remain in training a neural network using Back-Propagation? Explain with appropriate example. (10)
   d) What is the condition to use a function as Activation Function in neural network? Graphically represent two different forms of sigmoidal functions and their derivatives. Explain similarity and dissimilarity between the functions. (10)

3. a) What are the basic differences between Radial Basis Function (RBF) network and Multi Layer Perceptron (MLP)? Briefly describe training steps of RBF networks. (09)
   b) Why ensemble of neural networks is essential when an MLP solve any arbitrary function? What are the conditions for ensemble construction? (06)
   c) What are the different techniques for training individual networks in ensemble construction? Mention different ensemble methods for each type. (08)
   d) Describe an ensemble method which uses sampling of individual patterns. Is such technique suitable for ensemble of Decision Tree? Explain briefly. (12)

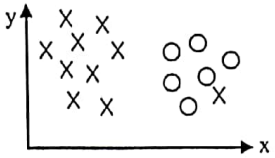
4. a) What do you mean by Swarm Intelligence (SI)? Briefly describe common features of SI based algorithms. (05)
   b) Explain importance of exploration and exploitation properties in SI algorithms. Describe exploration and exploitation matter in Particle Swarm Optimization (PSO). (10)
   c) In Ant Colony Optimization (ACO), how an ant complete a tour in solving Travelling Salesman Problem (TSP)? (08)
   d) What do you understand by the terms Global Search, Local Search and Blind Search? Does GA follows the search methods? Explain your justification with techniques included in GA. (12)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What are support vectors in Support Vector Machine (SVM)? How does Linear SVM work? Explain. (10)
   b) How can SVM classify non-linearly separable data? Explain with an example and proper figure. (07)
   c) Can SVM be over trained? What can be the effect of over training? How will you detect over fitting of a SVM model? (08)

d) Can you classify the data points (x, 0) depicted in the following figure using LSVM?



Explain the procedure of finding the correct hyper plane for the data points.

6. a) Define "Occam's Razor Principle". Explain its effect on Decision Tree. (07)
   b) "Entropy measures the impurity of training data instances"-is this statement true or false? Explain with proper reasoning. (08)
   c) Formulate a decision tree based on the following experience table where Buys-computer is set as target class. (20)

Records	Age	Income	Student	Credit-Rating	Buys-computer
1	<= 30	High	No	Fair	No
2	<= 30	High	No	Excellent	No
3	31 … 40	High	No	Fair	Yes
4	> 40	Medium	No	Fair	Yes
5	> 40	Low	Yes	Fair	Yes
6	> 40	Low	Yes	Excellent	No
7	31 … 40	Low	Yes	Excellent	Yes
8	<= 30	Medium	No	Fair	No
9	= 30	Low	Yes	Fair	Yes
10	> 30	Medium	Yes	Fair	Yes
11	<= 30	Medium	Yes	Excellent	Yes
12	31 … 40	Medium	No	Excellent	Yes
13	31 … 40	High	Yes	Fair	Yes
14	> 40	Medium	No	Excellent	No

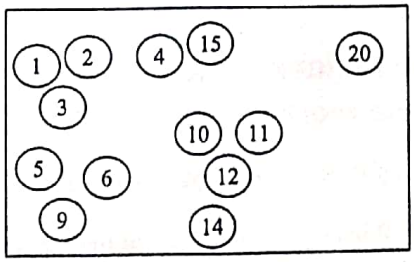
7. a) Why Pre-pruning and Post-pruning are necessary for decision trees? How does error propagate in decision tress? (10)
   b) How do you tackle the curse of dimensionality? (05)
   c) Write short notes on: (i) Data condensation, and (ii) Data squashing. (08)
   d) How does Principle Component Analysis construct the principle components? Point out the steps briefly with an example. (12)

8. a) Consider the data in the following table: (15)

ID	Height	Age	Weight
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	5.5	38	??

Predict the weight of ID 11 based on their height and age using KNN.

   b) What are the strengths of K-means algorithm? (05)
   c) Differentiate between divisive clustering and agglomerative clustering. (08)
   d) Draw a dendrogram from the following figure: (07)



KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 1st Term Examination, 2018
Department of Computer Science and Engineering
CSE 4111
Machine Learning

TIME: 3 hours FULL MARKS: 210

N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Machine Learning? Briefly discuss some practical applications of Machine Learning. (05)
   b) Explain McCulloch-Pitts model of artificial neuron with appropriate sketch. Describe significance of its individual components. (08)
   c) Explain the difficulties of solving XOR problem with single neuron. Solve XOR problem through manipulation by hidden neurons for AND and OR logic. (10)
   d) How does the cross-validation technique evaluate machine learning algorithms? Explain. (06)
   e) "The choices of training data and target concepts are two important issues in designing a machine learning system"– justify the statement with appropriate examples. (06)

2. a) What is the key difference between traditional algorithms and machine learning algorithms? (05)
   b) Write down the back propagation (BP) weight update rule for both hidden layer neurons and output layer neurons where sigmoidal and tanh functions are used as activation function in output and hidden layers, respectively. (10)
   c) How momentum parameter speed up training in BP? (06)
   d) What are MLP and RBF network? Write down the differences between MLP and RBF network. (06)
   e) Draw sketch diagram of a RBF network and put parameter values to solve XOR problem. (08)

3. a) Why ensemble of several neural networks is necessary when a network with hidden layer may approximate any arbitrary function? (05)
   b) What do you mean by independent, sequential and simultaneous training of component networks in ensemble construction? Mention prominent ensemble methods with each type of component network training. (10)
   c) Describe Bagging and AdaBoost ensemble methods with graphical representation. Compare the methods in different issues such as, training set size, ensemble decision and so on. (10)
   d) Describe an ensemble method which promotes diversity altering error of individual network. Is such technique suitable for ensemble of Decision Tree? Explain. (10)

4. a) What do you understand by Swarm Intelligence (SI)? Write down common features of SI based algorithms. (05)
   b) What do you mean by exploration and exploitation? Explain exploration and exploitation tradeoff in Particle Swarm Optimization (PSO). (08)
   c) "In PSO, each and every particle changes position in each iteration even the particle is the global best (G)"– are you agree with the statement? Justify your answer with appropriate assumptions in velocity update equation. (08)
   d) Write down the equations of transition probability and pheromone update of Ant Colony Optimization (ACO). Describe effects of key parameters in ACO. (09)
   e) Describe significance properties of ACO with respect to other population based algorithms. (05)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Define GA. Explain the working principle of GA with a flowchart. (07)
   b) "Crossover is explorative and Mutation is exploitative"– justify the statement with respect to GA. What are the effects of too high and too low Mutation rate? (10)
   c) Write down the similarities and dissimilarities between GA and PSO. (06)
   d) Apply the Enhanced Edge Recombination algorithm on the following two chromosomes to produce two child. Show all steps. (12)

Parent 1:	4	2	1	3	5	6
Parent 2:	4	3	1	2	6	5

6. a) What is the problem of using normal crossover and mutation operators in GA for solving Travelling Salesman Problem (TSP)? Briefly explain Insert Mutation operator for TSP. (
   b) What is Decision Tree? Write down the steps of Decision Tree construction. (07)
   c) What is Information Gain? How does Information Gain handle continuous-valued attributes? (07)
   d) Consider the following table where attribute 'Status' represents class label. Find the root attribute of the Decision Tree using Gain Ratio as the attribute selection measure. (12)

Department	Age	Status	Salary
Sales	31··35	Senior	Medium
Sales	26··30	Junior	Low
Marketing	36··50	Senior	High
Secretary	26··30	Senior	Medium
Secretary	36··50	Senior	High
Sales	36··50	Junior	High
Marketing	31··35	Senior	Medium
Marketing	26··30	Junior	Low
Secretary	31··35	Junior	Medium
Secretary	26··30	Junior	Low

7. a) Distinguish between Supervised and Unsupervised learning. (06)
   b) What is the problem associated with K-means clustering algorithm? How K-mediod clustering method solve this problem? Explain. (12)
   c) What are the advantages of Hierarchical clustering over Partitioning method of clustering? (05)
   d) Consider the following 5 data points. (12)

   (1, 1), (3, 2), (7, 9), (8, 7), (9, 8)

   Find the clusters using Agglomerative clustering and use Complete Link proximity as distance measure. Show all steps.

8. a) What is the problem of DBSCAN algorithm? Discuss. (05)
   b) Explain core-distance and reachability-distance with graphical representation. (06)
   c) Briefly explain the curse of dimensionality problem. (06)
   d) Consider the following dataset with 2 dimensions. (18)

Dimensions 1	Dimensions 2
2.5	2.4
0.5	0.7
2.2	2.9
1.9	2.2
2.5	3.0
2.7	2.5
2.0	1.6
1.0	1.1
2.5	2.6
2.1	1.9

Using Principal Component Analysis (PCA), reduce the dataset to one dimension. Explain each step with necessary calculations.

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2017
Department of Computer Science and Engineering
CSE 4211
Machine learning

TIME: 3 hours FULL MARKS: 210

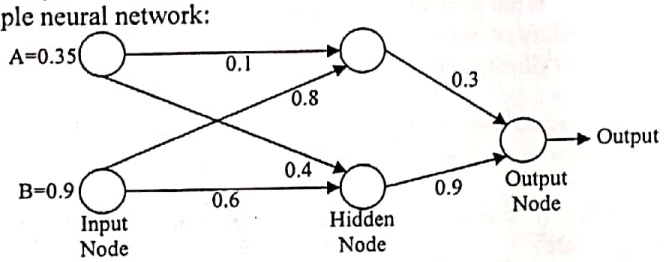
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

(Answer **ANY THREE** questions from this section in Script A)

1. a) What is Machine Learning? Briefly discuss some practical applications of Machine Learning. (08)
   b) Distinguish between supervised and unsupervised learning with example. (07)
   c) How do you differentiate regression and classification problems in Machine Learning? Give some practical examples. (10)
   d) Briefly explain the curse of dimensionality problem. (10)

2. a) What do you mean by feed forward network? (05)
   b) Consider the simple neural network: (10)



   (i) Perform a forward pass on the network.
   (ii) Perform a backward pass on the network
   (iii) Perform a further forward pass and comment on the result.
   Use sigmoid function as activation function.
   c) Explain in difficulties to solve XOR problem with single neuron. Solve XOR problem through manipulation by hidden neurons for AND and OR logic. (10)
   d) Draw minimal neural models for the following two logics having inputs A, B and C. (10)
   (i) NN1=(A+B)C' (ii) NN2=ABC'

3. a) What is the effect of learning rate in BP? How momentum parameters speed up training in BP? (10)
   b) What are the basic differences between RBF network and MLP. (08)
   c) What is local gradient in BP training procedure? For a neural network with two hidden layers, write local gradients for each layer. (10)
   d) Differentiate learning and generalization? Which one is more desirable and why? (07)

4. a) What is the purpose of mathematical optimization? Define following terminologies with respect to mathematical optimization: (i) Objective function (ii) variables/parameter (iii) Constraints (iv) Constrained and unconstrained optimization (v) language multiplier. (07)
   b) How to solve an unconstrained optimization problem using Gradient descent. Explain in detail. (08)
   c) Solve the following constrained optimization problems: (i) Given the objective function (20)

$f(x,y) = x^3 + y^3$ *subject* to

$g_1(x,y) = x^2 - 1 \geq 0$

$g_2(x,y) = y^2 - 1 \geq 0$

Find the optimum value.

(ii) A company has determined that its production is the cobb-douglas function, $f(x,y) = x^{2/3}y^{1/3}$ where x=number of labour hours, y=number of capital units, the budget constraint for the company is 100X+100Y=400000. What is the maximum production?

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) What is meant by Bio-inspired computing? Write some the popular Bio-inspired computing algorithm. (10)
   b) Why do we prefer heuristic approaches over classical methods to solve optimization problems? Explain with example. (10)
   c) Explain the trade-off criteria for deciding whether to use a heuristic for solving a given problem. (08)
   d) Define Genetic Algorithm (GA). Write simple GA. Write nature to computer mapping. (07)

6. a) Explain exploitation and exploration with examples. (10)
   b) Why do we not use one point crossover to solve TSP? Explain. (07)
   c) State the differences between roulette wheel selection and tournament selection. (08)
   d) Solve traveling salesman problem using particle swarm optimization. (10)

7. a) What is Swarm Intelligence (SI)? What are the properties of SI system? (10)
   b) Write PSO algorithm. What problems are arise when PSO is used in multimodal problems? (10)
   c) Write working procedure of MPSO. (05)
   d) Differentiate between Gbest swarm and Lbest swarm. (10)

8. a) Describe the important of transition probability in ACO algorithm. (10)
   b) "Pheromone evaporation pushes ACO convergence." is it true? Justify your opinion. (07)
   c) Write ACO algorithm. (08)
   d) Explain the employed bee phase, on looker bee phase and scout bee phase in ABC optimization algorithm. (10)

KHULNA UNIVERSITY OF ENGINEERING & TECHNOLOGY
B.Sc. Engineering 4th Year 2nd Term Examination, 2015
Department of Computer Science and Engineering
CSE 4211
Machine Learning

TIME: 3 hours FULL MARKS: 210

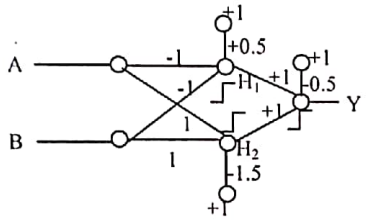
N.B. i) Answer **ANY THREE** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

SECTION A

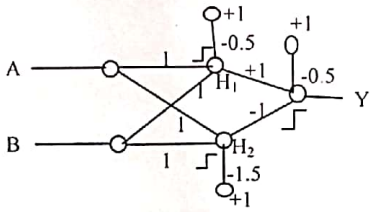
(Answer **ANY THREE** questions from this section in Script A)

1. a) Explain McCulloch-Pitts model of artificial neuron with appropriate sketch. Describe significance of its individual components. (09)
   b) Describe importance of bias weight in artificial neuron. "Only changing the weight value of bias operation of a neuron may change"-justify the statement for appropriate logic gates. (09)
   c) Draw single neuron model for following logics: (08)
   i) $y = x_1' x_2'$, ii) $y = x_1' x_2$, iii) $y = x_1 x_2$, and iv) $y = (x_1 x_2)'$.
   d) Describe shortly the three fundamental and practical issues of learning theory. (09)

2. a) Look the following two neural networks model. (10)



(i)



(ii)

For the model tabulate the outputs of H_1, H_2 and Y for inputs (A and B) values 0 or 1. Are the model performs any logic gate operation? Which ones?
   b) What is the condition to use a function as activation function in ANN? Graphically represents two different forms sigmoidal functions and their derivatives. How derivatives indicate the stability of ANN training? (10)
   c) What is the naming significance of "Back Propagation (BP)" algorithm? Describe basic steps of BP. (07)
   d) What is local gradient in BP training procedure? For an ANN with two hidden layers, write local gradients of each layer. (08)

3. a) What is the effect of learning rate in BP? How momentum parameters speed up training in BP? (05)
   b) What are basic differences between Radial Basic Function (RBF) network and MLP? (05)
   c) Why hybrid training algorithm is required for training RBF networks? Draw sketch diagram of a RBF network and put parameter values to solve XOR problem. (08)
   d) What are the properties and usefulness of the followings: (07)
   i) Local Addressable Memory, ii) Content Addressable Memory, and iii) Associative Memory.
   e) Calculate the weight set (w) of a Hetero- Associative Memory for the following samples: (10)

Pattern	Source, s(p)	Target, t(p)
1	(1 0 0 0)	(1 0)
2	(1 1 0 0)	(1 0)
3	(0 0 0 1)	(0 1)
4	(0 0 1 1)	(0 1)

Finally verify that the system work correctly.

4. a) Differentiate learning and generalization. Which one is more desirable and why? (05)
   b) Why ensemble of classifiers is necessary when a MLP able to capture any arbitrary function? What is the condition for better ensemble construction? (05)
   c) Among the studied ensemble methods, whose are able to take facility of multi-processing system? How? (07)

d) Describe an ensemble method which promotes diversity through artificial patterns. (08)
e) What is support vectors in support vector machine (SVM)? How SVM may classify non linearly separable data? Explain with appropriate examples. (10)

SECTION B

(Answer **ANY THREE** questions from this section in Script B)

5. a) Why we prefer nature-inspired heuristic methods although they might not give exact solution for a particular optimization task? (05)
b) Why selection is necessary in Genetic Algorithm (GA)? Write down the differences between Rank based selection and Tournament selection. (12)
c) Write down the importance of using mutation in GA. Also write down the effects of mutation rate in GA. (07)
d) Is it possible to find out the population size from the Roulette wheel given below? Explain. (06)



Fig. 5(d)

e) What is Elitism? Why is it important in GA? (05)

6. a) "In Particle Swarm Optimization (PSO) every particle updates position in each and every iteration even it is the global best"-justify the statement from the velocity and position update equations. (08)
b) Write down the working procedure of Attractive Repulsive PSO and Multiswarm PSO. (08)
c) What is the effect of Lifespan controller in ALC-PSO? How Lifespan of a leader is adaptively tuned? Explain. (12)
d) Write down the similarities and dissimilarities between GA and PSO. (07)

7. a) Describe the importance of transition probability in selecting the next city in TSP using Ant Colony Optimization (ACO). (10)
b) "Pheromone evaporation pushes ACO convergence"-is it true? Justify your opinion. (08)
c) Explain the Employed bee phase, onlooker bee phase and scout bee phase of Artificial Bee Colony Optimization (ABCO) algorithm. (12)
d) Explain differences of Biogeography-Based Optimization (BBO) from ACO and PSO. (05)

8. a) How does Cuckoo Search (CS) algorithm provide exploration and exploitation? Explain. (08)
b) How does Bacterial Foraging Optimization Algorithm (BFOA) find the optimum solution? (05)
c) What is the benefit of using OPTICS clustering algorithm? How this algorithm works? Explain. (08)
d) Define the core point and border point. Consider the data points in the following figure Fig. 8(d) with radius where Minpts = 6. Find out the core points and border points. (06)

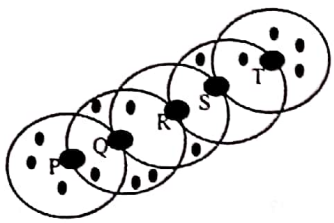


Fig. 8(d)

Is P and T density reachable from each other?

e) Consider a set of records that have three attributes namely salary, age and vacation. How CLIQUE algorithm clusters data objects? Explain. (08)